

Penetration Test Report

AI-Powered Windows Security Assessment

HIGH RISK

Target IP	172.20.10.5
Hostname	N/A
OS Detected	Windows
Scan Date	May 17, 2026 — 10:13
Duration	12.24 seconds
Open Services	3
CVEs Matched	12
Risk Score	6.6 / 10 — HIGH RISK

This report was generated by the AI-Powered Windows Penetration Testing Assistant. For authorized security assessment use only.

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01 — Executive Summary

This assessment identified **3 open service(s)** on the target system and matched **12 CVE vulnerabilities** from the local Windows CVE database. Overall risk: **HIGH RISK**. Risk score: **6.6/10**.

Of the 12 matched CVEs, 2 are recent (post-2020) and may be unpatched, while 10 are older CVEs likely patched on modern Windows systems.

Severity	Count	Priority
CRITICAL	2	Immediate action required
HIGH	7	Remediate within 24-48 hours
MEDIUM	3	Remediate within 30 days
LOW	0	Remediate when possible

AI Assessment Overview

Executive Summary Overall Risk Assessment: The target, a Windows system with open ports for RPC and SMB services, is highly vulnerable due to multiple critical vulnerabilities across both protocols. Specifically, the 135/RPC port contains three high-severity vulnerabilities (CVE-2020-1113, CVE-2017...

02 — Scan Information

Target IP	172.20.10.5
Hostname	N/A
OS Detected	Windows
Scan Date	2026-05-17 10:13:13
Duration	12.24 seconds
Scanner	Nmap 7.98
Command	/usr/lib/nmap/nmap --privileged -vvv -p 135,139,445 -4 -sV -sC --script smb-vuln...

03 — Open Ports & Services

Port	Protocol	Service	Product / Banner
135	TCP	RPC	Microsoft Windows RPC
139	TCP	SMB	Microsoft Windows netbios-ssn
445	TCP	SMB	microsoft-ds?

04 — CVE Vulnerability Findings

CVE matches are based on service detection from the local NIST NVD dataset. Patch status is estimated by publication date.

Service: 135/rpc

CVE ID	CVSS	Severity	Description	Patch Status
CVE-2016-0178	8.8	HIGH	The RPC NDR Engine in Microsoft Windows Vista SP2, Windows Server 2008 SP2 and R2 SP1, Win...	Likely Patched

CVE-2017-0100	7.8	HIGH	A DCOM object in Helppane.exe in Microsoft Windows 7 SP1; Windows Server 2008 R2; Windows ...	Likely Patched
CVE-2020-1113	7.5	HIGH	A security feature bypass vulnerability exists in Microsoft Windows when the Task Schedule...	May be Unpatched
CVE-2016-0128	6.8	MEDIUM	The SAM and LSAD protocol implementations in Microsoft Windows Vista SP2, Windows Server 2...	Likely Patched

Service: 139/smb

CVE ID	CVSS	Severity	Description	Patch Status
CVE-2017-0021	9.0	CRITICAL	Hyper-V in Microsoft Windows 10 1607 and Windows Server 2016 does not properly validate vS...	Likely Patched
CVE-2015-2474	9.0	CRITICAL	Microsoft Windows Vista SP2 and Server 2008 SP2 allow remote authenticated users to execut...	Likely Patched
CVE-2017-0144	8.8	HIGH	The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo...	Likely Patched
CVE-2017-0145	8.8	HIGH	The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo...	Likely Patched
CVE-2022-24500	8.8	HIGH	Windows SMB Remote Code Execution Vulnerability	May be Unpatched
CVE-2017-0146	8.8	HIGH	The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo...	Likely Patched
CVE-2018-8444	5.9	MEDIUM	An information disclosure vulnerability exists in the way that the Microsoft Server Messag...	Likely Patched
CVE-2017-0276	5.9	MEDIUM	Microsoft Server Message Block 1.0 (SMBv1) allows an information disclosure vulnerability ...	Likely Patched

05 — AI Analysis & Recommendations

Analysis generated by Qwen2.5:3b AI model using RAG from local CVE dataset.

Executive Summary

Overall Risk Assessment: The target, a Windows system with open ports for RPC and SMB services, is highly vulnerable due to multiple critical vulnerabilities across both protocols. Specifically, the 135/RPC port contains three high-severity vulnerabilities (CVE-2020-1113, CVE-2017-0144, CVE-2016-0128), and the 139/SMB port has eight critical vulnerabilities (CVE-2017-0021, CVE-2015-2474, CVE-2022-

Most Critical Vulnerabilities Identified:

- CVE-2020-1113 (Task Scheduler Security Feature Bypass)
- CVE-2017-0021 (Hyper-V System Data Structure Corruption)

Vulnerability Analysis

Rank by Exploitability and Impact

CVE ID | Severity | Published | Patch Status

CVE-2020-1113 | HIGH | 2020-05-21 | ✓ Recent CVE — may still be unpatched

CVE-2017-0021 | CRITICAL | 2017-03-17 | ■■■ Likely patched on modern Windows 10/11 (published pre-2020)

CVE-2020-1113:

This vulnerability allows attackers to execute arbitrary code on a target OS, making it highly critical. The Task Scheduler service is a core part of the operating system and can be used for remote code execution.

CVE-2017-0021:

The Hyper-V server in Windows 1607 and Server 2016 does not properly validate vSMB packet data, which allows attackers to execute arbitrary code. This is a critical vulnerability that impacts the virtualization layer of the operating system.

Attack Vectors

Most Realistic Attack Paths

CVE-2020-1113 (Task Scheduler Security Feature Bypass):

1. Exploit Method: The attacker would exploit this vulnerability by sending crafted packets to a vulnerable Task Scheduler service.
2. Metasploit Module: msfvenom can be used to generate the malicious payload, and msfconsole for exploitation.

3. Steps:

- Identify the vulnerable Task Scheduler service using tools like Nmap or similar scanning software.
- Craft a malicious payload that exploits CVE-2020-1113.
- Execute the payload on the target system.

CVE-2017-0021 (Hyper-V System Data Structure Corruption):

1. Exploit Method: The attacker would send crafted packets to the SMBv1 server, which is vulnerable due to this CVE.
2. Metasploit Module: Use msfconsole for exploitation and Metasploit modules like msfvenom for crafting payloads.

3. Steps:

- Identify the SMBv1 server using tools like Nmap or similar scanning software.
- Craft a malicious payload that exploits CVE-2017-0021.
- Execute the payload on the target system.

Recommended Tools & Techniques

Metasploit Modules

CVE ID | Metasploit Module

CVE-2020-1113 | exploit/windows/smb/ms16_058_sam

CVE-2017-0021 | exploit/multi/handler (for testing)

Other Tools

- Nmap: For scanning and identifying vulnerable services.
- Burp Suite: For intercepting and analyzing SMB traffic.

Remediation Steps

CVE-2020-1113:

- Apply the latest security patches provided by Microsoft for Windows 10/11. Ensure that all relevant updates are installed to mitigate this vulnerability.

CVE-2017-0021:

- Upgrade to a more recent version of Hyper-V, such as Windows Server 2019 or later, which includes the necessary security patches.
- Alternatively, apply the Microsoft Security Update KB4538465 for Windows Server 2016.

Risk Score

Overall Score: 8/10

Justification: The target is highly vulnerable to multiple critical vulnerabilities. CVE-2020-1113 and CVE-2017-0021 are the most severe, with CVE-2020-1113 having a CVSS score of 8.8 and CVE-2017-0021 having a CVSS score of 9.0. These vulnerabilities have significant real-world exploitability and impact.

Conclusion

The target system is highly vulnerable to multiple critical vulnerabilities, including the Task Scheduler Security Feature Bypass (CVE-2020-1113) and Hyper-V System Data Structure Corruption (CVE-2017-0021). Immediate remediation should be prioritized to mitigate these risks.

06 — Remediation Steps

RPC	Restrict RPC via firewall. Apply all Windows security patches. Disable unused RPC services.
SMB	Disable SMBv1. Apply MS17-010 patch (KB4012212). Block port 445 externally. Enable SMB signing.

General Security Hardening

- Apply all pending Windows security updates via Windows Update
- Enable Windows Defender and keep definitions up to date
- Implement network segmentation and strict firewall rules

- Use principle of least privilege for all accounts
- Enable Windows Event Logging and monitor for suspicious activity
- Implement multi-factor authentication (MFA) for all remote access

07 — Risk Score

Overall Risk Score	6.6 / 10
Risk Level	HIGH RISK
Total CVEs	12
Critical	2
High	7
Medium	3
Low	0
Recent CVEs	2
Likely Patched	10

Risk score is calculated based on recent CVEs (post-2020) only. Older CVEs are weighted lower as they are likely patched on modern Windows systems.

08 — Legal Disclaimer

This report was prepared exclusively for the authorized recipient. All testing was conducted with explicit written authorization from the system owner.

Unauthorized access to computer systems is a criminal offence under the Computer Misuse Act 1990 (UK), Computer Fraud and Abuse Act (USA), and equivalent legislation worldwide.

CVE data sourced from the National Vulnerability Database (NVD) — NIST (<https://nvd.nist.gov/>). AI analysis by Qwen2.5:3b via Ollama (RAG mode — local dataset only).

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